

Lab 05 - Expected evidence

Digests, runtime versions and sizes are cluster-specific. Structure and marker strings are what you compare against.

— — —

1. The runtime behind the CRI socket

```
$ kubectl get nodes -o wide
```

NAME	STATUS	ROLES	AGE	VERSION	INTERNAL-IP
EXTERNAL-IP	OS-IMAGE		KERNEL-VERSION		CONTAINER-RUNTIME
kcna-qa-20260905-control-plane	Ready	control-plane	102m	v1.37.0	172.18.0.2
<none>	Debian GNU/Linux	12 (bookworm)	6.10.14-linuxkit		containerd://2.1.4

```
$ kubectl get nodes -o jsonpath='{.items[0].status.nodeInfo.containerRuntimeVersion}'{"\n"}'
containerd://2.1.4
```

2. The node's image store, straight from the Node object

```
$ kubectl get nodes -o jsonpath='{range .items[0].status.images[*]}{.sizeBytes}{"\t"}{.names[0]}{"\n"}{end}' | sort -rn | head -n 6
147718971██registry.k8s.io/etc@sha256:
e78e30b6b5c2ea9d2b0f8b1a2f9e1bd5e9e9b3f18ea9d1c0b2b74f6c3e2e9b40d2
92632544██registry.k8s.io/kube-apiserver@sha256:
1a1f05a2cd7c2b2e3e0fa45c7c7f74a4e3fdc7fbc25da8b0e0d84a8b7f6cf12
21674089██docker.io/library/nginx@sha256:
3d696e8357051647b844d8c7cf4a0aa71e84379999a4f6af9b8ca1f7919ade42
4264178██docker.io/library/alpine@sha256:
beefdbd8a1da6d2915566fde36db9db0b524eb737fc57cd1367effd16dc0d06d
4042190██docker.io/library/busybox@sha256:
2376a0c12759aa1214ba83e771ff252c7b1663216b192fbe5e0fb364e952f85c
744000██registry.k8s.io/pause@sha256:
7031c1b283388d2c2e09b57badb803c05ebd362dc88d84b480cc47f72a21097
```

Every entry is repository@sha256:.... The node's store is content addressed; tags are just labels that point into it.

3. Tag-pinned Pod: asked-for vs resolved

```
$ kubectl -n kcna-lab05 get pod image-catalog
NAME          READY   STATUS    RESTARTS   AGE
image-catalog 1/1     Running   0           24s

$ kubectl -n kcna-lab05 get pod image-catalog -o jsonpath='{.spec.containers[0].image}'
nginx:1.27-alpine

$ kubectl -n kcna-lab05 get pod image-catalog -o jsonpath='{.status.containerStatuses[0].image}'
docker.io/library/nginx:1.27-alpine

$ kubectl -n kcna-lab05 get pod image-catalog -o jsonpath='{.status.containerStatuses[0].imageID}'
docker.io/library/nginx@sha256:3d696e8357051647b844d8c7cf4a0aa71e84379999a4f6af9b8ca1f7919ade42
```

Three different strings for one image:

Field	Value	Meaning
spec....image	nginx:1.27-alpine	what you wrote
status....image	docker.io/library/nginx:1.27-alpine	what the reference normalises to
status....imageID	docker.io/library/nginx@sha256:3d69...	what the runtime actually resolved and ran

4. Datasets served

```
$ kubectl -n kcna-lab05 exec image-catalog -c web -- wget -qO- http://127.0.0.1/image-inventory.csv | grep -v '^#'
component,image_ref,registry,repository,tag,pin_style,approx_size_mb,base,notes
depot-web,nginx:1.27-alpine,docker.io,library/nginx,1.27-alpine,tag,22,alpine,static content only
depot-api,busybox:1.36,docker.io,library/busybox,1.36,tag,2,scratch+busybox,PoC stand-in for the Go API
depot-jobs,alpine:3.20,docker.io,library/alpine,3.20,tag,4,alpine,batch and cron workloads
depot-sandbox,registry.k8s.io/pause:3.9,registry.k8s.io,pause,3.9,tag,1,scratch,pod sandbox infra container
depot-web-locked,nginx@sha256:<digest>,docker.io,library/nginx,none,digest,22,alpine,production pin - digest resolved at release time
```

```
$ kubectl -n kcna-lab05 exec image-catalog -c web -- wget -qO- http://127.0.0.1/reference-forms.tsv | grep -v '^#' | head -n 3
nginx█docker.io/library/nginx:latest█none - tag is mutable AND implicit
nginx:1.27-alpine█docker.io/library/nginx:1.27-alpine█weak - tag can be re-pointed
library/nginx:1.27-alpine█docker.io/library/nginx:1.27-alpine█weak - same as above
```

5. imagePullPolicy matrix

```
$ kubectl -n kcna-lab05 get pods -o custom-columns='NAME:.metadata.name,IMAGE:.spec.containers[0].image,POLICY:.spec.containers[0].imagePullPolicy,STATUS:.status.containerStatuses[0].state'
NAME          IMAGE                                POLICY
STATUS
image-catalog  nginx:1.27-alpine                    IfNotPresent
map[running:...]
pp-always      busybox:1.36                         Always
map[running:...]
pp-ifnotpresent busybox:1.36                         IfNotPresent
map[running:...]
pp-never       meridianfreight.example.com/depot-portal/depot-api:0.1.0-poc Never
map[waiting:map[messagge:Container image "... is not present with pull policy of Never reason: ErrImageNeverPull]]
```

```
$ kubectl -n kcna-lab05 get pod pp-never
NAME      READY   STATUS             RESTARTS   AGE
pp-never  0/1     ErrImageNeverPull  0           19s

$ kubectl -n kcna-lab05 describe pod pp-never | tail -n 5
Events:
  Type    Reason            Age   From              Message
  ----    -
  Normal  Scheduled         21s   default-scheduler Successfully assigned kcna-lab05/pp-never to kcna-qa-20260905-control-plane
  Warning ErrImageNeverPull 6s (x3 over 21s) kubelet           Container image "meridianfreight.example.com/depot-portal/depot-api:0.1.0-poc" is not present with pull policy of Never
```

There is no Pulling event - with Never the kubelet does not contact a registry at all.

Events for the two working policies:

```
$ kubectl -n kcna-lab05 describe pod pp-always | grep -E 'Pulling|Pulled'
Normal Pulling 30s kubelet Pulling image "busybox:1.36"
Normal Pulled 28s kubelet Successfully pulled image "busybox:1.36" in 1.842s (1.842s
including waiting). Image size: 4042190 bytes.

$ kubectl -n kcna-lab05 describe pod pp-ifnotpresent | grep -E 'Pulling|Pulled'
Normal Pulled 30s kubelet Container image "busybox:1.36" already present on machine
```

already present on machine with no preceding Pulling is the whole difference.

6. Failure injection - stale digest

```
$ kubectl apply -f manifests/20-digest-pinned.yaml
pod/depot-web-locked created

$ kubectl -n kcna-lab05 get pod depot-web-locked
NAME                READY   STATUS    RESTARTS   AGE
depot-web-locked    0/1     ErrImagePull  0           7s
```

```
$ kubectl -n kcna-lab05 describe pod depot-web-locked | tail -n 8
Events:
  Type      Reason      Age           From          Message
  ----      -
  Normal    Scheduled   48s           default-scheduler Successfully assigned kcna-lab05/
depot-web-locked to kcna-qa-20260905-control-plane
  Normal    Pulling     18s (x3 over 48s) kubelet        Pulling image "nginx@sha256:
9c1b5b6f0f6cc1e7d0d2a3c4e5f60718293a4b5c6d7e8f90a1b2c3d4e5f60718"
  Warning   Failed     17s (x3 over 47s) kubelet        Failed to pull image "nginx@sha256:
9c1b5b6f...": failed to pull and unpack image "docker.io/library/nginx@sha256:9c1b5b6f...":
failed to resolve reference "docker.io/library/nginx@sha256:9c1b5b6f...": docker.io/library/
nginx@sha256:9c1b5b6f...: not found
  Warning   Failed     17s (x3 over 47s) kubelet        Error: ErrImagePull
  Normal    BackOff    3s (x4 over 46s) kubelet        Back-off pulling image
"nginx@sha256:9c1b5b6f..."
  Warning   Failed     3s (x4 over 46s) kubelet        Error: ImagePullBackOff
```

Repair using the digest this cluster resolved:

```
$ DIGEST=$(kubectl -n kcna-lab05 get pod image-catalog -o jsonpath='{.status.containerStatuses[
0].imageID}')
$ echo "$DIGEST"
docker.io/library/nginx@sha256:3d696e8357051647b844d8c7cf4a0aa71e84379999a4f6af9b8ca1f7919ade42

$ kubectl -n kcna-lab05 set image pod/depot-web-locked "web=nginx@${DIGEST##*@}"
pod/depot-web-locked image updated

$ kubectl -n kcna-lab05 get pod depot-web-locked
NAME                READY   STATUS    RESTARTS   AGE
depot-web-locked    1/1     Running   0           2m11s

$ kubectl -n kcna-lab05 get pod depot-web-locked -o jsonpath='{.spec.containers[0].image}{"\n"}{.status.containerStatuses[0].imageID}{"\n"}'
nginx@sha256:3d696e8357051647b844d8c7cf4a0aa71e84379999a4f6af9b8ca1f7919ade42
docker.io/library/nginx@sha256:3d696e8357051647b844d8c7cf4a0aa71e84379999a4f6af9b8ca1f7919ade42
```

With a digest pin, spec.image and status.imageID name the same content. That is the definition of a reproducible deployment.

7. Verification script

```
$ bash verification/checks.sh

== 0. Cluster reachability
[PASS] kubectl can reach the API server

== 1. Namespace and catalog ConfigMap built from data/
[PASS] namespace kcna-lab05 exists
[PASS] ConfigMap image-catalog carries image-inventory.csv
[PASS] ConfigMap image-catalog carries reference-forms.tsv

== 2. The runtime this node actually uses (CRI)
[PASS] node reports a CRI runtime: containerd://2.1.4

== 3. Tag-pinned catalog Pod
[PASS] Pod image-catalog is Ready
[PASS] spec.containers[0].image is the TAG you asked for (nginx:1.27-alpine)
[PASS] status.imageID resolved to a content-addressed digest
[PASS] digest is a well-formed sha256:<64 hex> string

== 4. The dataset is really being served
[PASS] GET /image-inventory.csv returns the Meridian inventory
[PASS] GET /reference-forms.tsv returns the reference normalisation card

== 5. Governance: running images vs the approved inventory
[PASS] image-catalog runs an image listed in data/image-inventory.csv
[PASS] pp-never runs an unapproved image (meridianfreight.example.com/depot-portal/depot-api:
0.1.0-poc) - drift is detectable from data/

== 6. imagePullPolicy matrix
[PASS] pp-always declares imagePullPolicy: Always
[PASS] pp-ifnotpresent declares imagePullPolicy: IfNotPresent
[PASS] pp-never declares imagePullPolicy: Never
[PASS] pp-never is ErrImageNeverPull - the kubelet never contacted a registry

== 7. Digest-pinned Pod
[PASS] depot-web-locked is pinned by digest, not by tag
[PASS] depot-web-locked is Ready - you substituted a digest this cluster can resolve
[PASS] the repaired digest matches image-catalog's resolved digest exactly

Result: 19 passed, 0 failed
```

If you run the script before repairing depot-web-locked, section 7 prints [SKIP] depot-web-locked is still ImagePullBackOff - you are mid-way through section 6 and the total is 17 passed, 0 failed.